

Deep Generative Models

Chapter 5: Variational Inference and Variational Autoencoding

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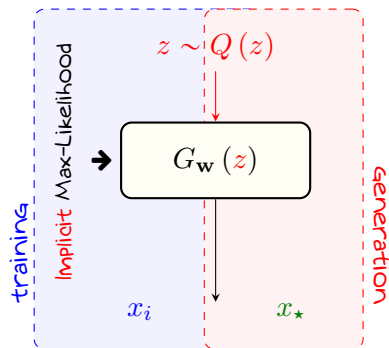
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Summer 2026

Modern Data Generation

Recall that we are looking into modern generative models

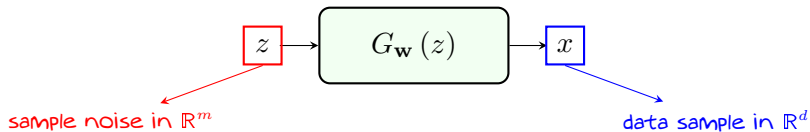
modern generative models mainly learn *how to sample!*



Recall: Generic Model for Generator

Generator Model

Generator model $G_{\mathbf{w}} : \mathbb{R}^m \mapsto \mathbb{R}^d$ is a mapping that maps a **latent sample** $z \sim P(z)$, typically Gaussian noise, into a data sample

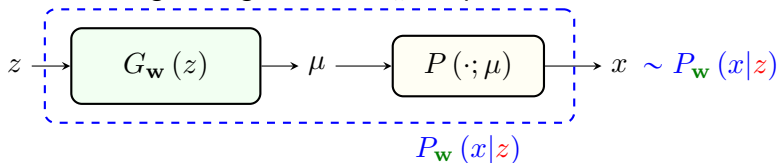


GANs consider **deterministic** generators

- + Can we also consider a **probabilistic** generator?!
- Sure! Why not?!
- + But how does it look like?
- Let's build one together!

Modeling Probabilistic Generator

Consider following setting: a model $G_{\mathbf{w}}$ computes some statistics from *latent*



We use the statistics to specify a distribution $P(x; \mu)$ and sample from it

↳ Sample x is drawn as

$$x \sim P(x; \mu) = P(x; G_{\mathbf{w}}(z))$$

↳ The distribution depends on *model parameters* $\mathbf{w}$ and is conditioned on z

$$x \sim P(x; \mu) \equiv P_{\mathbf{w}}(x|z)$$

Probabilistic Generator: Gaussian Example I

Let's look at an example: say $G_{\mathbf{w}} : \mathbb{R}^m \mapsto \mathbb{R}^d$

- Sample a *latent* z and compute a *mean value*

$$\mu = G_{\mathbf{w}}(z)$$

- Set μ to be *the mean* of a Gaussian distribution and sample it
↳ with *identity covariance* it's very *easy to sample* this distribution

$$x \sim P(x; \mu) \equiv \mathcal{N}(\mu, 1) = \mathcal{N}(G_{\mathbf{w}}(z), 1)$$

We can think of this process as sampling x conditioned to z from

$$P_{\mathbf{w}}(x|z) = \frac{1}{(2\pi)^{d/2}} \exp \left\{ -\frac{\|x - G_{\mathbf{w}}(z)\|^2}{2} \right\}$$

Probabilistic Generator: Gaussian Example II

We can further extend this example: say $G_{\mathbf{w}} : \mathbb{R}^m \mapsto \mathbb{R}^d \times \mathbb{R}^{d \times d}$

- Sample a *latent* z and compute *mean and covariance*

$$\mu, \Sigma = G_{\mathbf{w}}(z)$$

- Sample a Gaussian distribution with *mean* μ and *covariance* Σ
 ↳ with *general covariance sampling* might need more computation

$$x \sim P(x; \mu, \Sigma) \equiv \mathcal{N}(\mu, \Sigma) = \mathcal{N}(G_{\mathbf{w}}(z))$$

We can think of this process as sampling from

$$P_{\mathbf{w}}(x|z) = \frac{1}{(2\pi|\Sigma_{\mathbf{w}}(z)|)^{d/2}} \exp \left\{ -\frac{(x - \mu_{\mathbf{w}}(z))^{\top} \Sigma_{\mathbf{w}}^{-1}(z) (x - \mu_{\mathbf{w}}(z))}{2} \right\}$$

Probabilistic Generation: General Form

We can formulate a general *latent-space probabilistic* generation as

Probabilistic Latent-Space Generation

To build a *probabilistic latent-space* generator

- 1 Sample *latent* space as $z \sim P(z)$
 - ↳ $P(z)$ is an *arbitrary* distribution, e.g., Gaussian
- 2 Compute *statistics* as $\mu = G_w(z)$
 - ↳ G_w is a *computational model*, e.g., NN
- 3 Sample data from $P(x; \mu) \equiv P_w(x|z)$
 - ↳ $P(x; \mu)$ is an arbitrary distribution *parameterized by μ*

- + Is such model *computationally* tractable?
- Let's check!

Probabilistic Generation: *Practical Setting*

Let's take a look at each step

- 1 Sample *latent* space as $z \sim P(z)$
 - ↳ $P(z)$ is an *arbitrary* distribution
 - ✓ We set $P(z)$ *simply to* $\mathcal{N}(0, 1)$
- 2 Compute *statistics* as $\mu = G_w(z)$
 - ✓ This is a simple forward pass
- 3 Sample from $P(x; \mu) \equiv P_w(x|z)$
 - ↳ $P(x; \mu)$ is an arbitrary distribution *parameterized by* μ
 - ✓ We set $P(x; \mu)$ *simply to* $\mathcal{N}(\mu, 1)$

Probabilistic Generation in Practice

In practice we work with *easy-to-simple distributions* $\equiv$ *Gaussian*

- + Wait a moment! Isn't output a *Gaussian sample*?! Isn't this *too limited*?!
 - *Careful!* The output is only *conditionally Gaussian*!

Model Distribution: *Example*

Consider a **scalar example**: say $z \sim \text{Unif}[0, 1]$ and $G_\theta : [0, 1] \mapsto \{0, 1\}$ is

$$\mu_\theta(z) = G_\theta(z) = \begin{cases} 0 & z < \theta \\ 1 & z \geq \theta \end{cases}$$

We set $P(x; \mu)$ to $\mathcal{N}(\mu, 1)$: we thus have

$$P_\theta(x|z) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - \mu_\theta(z))^2}{2} \right\}$$

This is though the distribution of x for a **given z** !

Distribution of x is determined by **marginalization** over **latent**

$$P_\theta(x) = \int P_\theta(x|z) P(z) dz$$

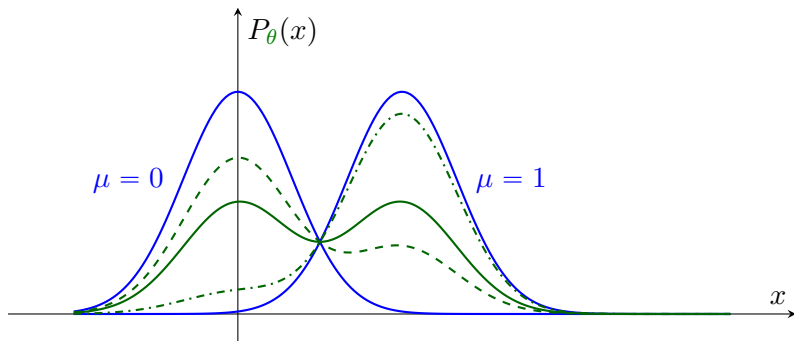
Model Distribution: *Example*

Let's use the *marginalization* to compute $P_{\theta}(x) \equiv$ *model* distribution

$$\begin{aligned}
 P_{\theta}(x) &= \int_0^1 P_{\theta}(x|z) P(z) dz \\
 &= \int_0^1 \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - \mu_{\theta}(z))^2}{2} \right\} \cdot 1 dz \\
 &= \int_0^{\theta} \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - 0)^2}{2} \right\} dz + \int_{\theta}^1 \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - 1)^2}{2} \right\} dz \\
 &= \frac{\theta}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2} \right\} + \frac{1 - \theta}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - 1)^2}{2} \right\}
 \end{aligned}$$

This is a *mixture* of *Gaussian distributions*

Model Distribution: *Example*



Say $\theta = 0.5 \rightsquigarrow$ the model distribution is average of the two Gaussians

↳ Setting $\theta = 0.7$ gives more weight to zero-mean Gaussian

↳ Setting $\theta = 0.1$ gives more weight to unit-mean Gaussian

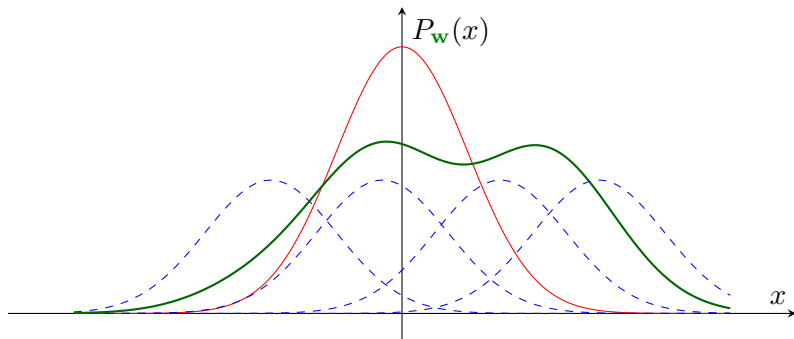
We can cover a class of distributions by changing θ

Model Distribution: *Mixture of Gaussians*

In practice, the model distribution is a mixture of *infinite Gaussians*

$$P_{\mathbf{w}}(x) = \int P_{\mathbf{w}}(x|z) P(z) dz$$

Here, $P(z)$ is Gaussian itself $\rightsquigarrow$ *we can make a large set of distributions*



Universal Model Distribution

Universality of Gaussian Mixtures (informal)

Consider the following distribution model on $\mathbb{X}$

$$P_{\mathbf{w}}(x) = \int P_{\mathbf{w}}(x|z) P(z) dz$$

whose conditional distributions $P_{\mathbf{w}}(x|z)$ are **learnable** Gaussians: this model describes a **dense** set of distributions on $\mathbb{X} \rightsquigarrow$ for almost any **smooth** $Q(x)$ there exists a **w** that

$$P_{\mathbf{w}}(x) \approx Q(x)$$

! Attention

We could **not** do this without **conditional** modeling $P_{\mathbf{w}}(x|z)$

Training via MLE

- + *Sounds good!* How can we then train this *probabilistic* model?!
- Well, we may try using **MLE**

The *likelihood* of *the model* for a given sample x is

$$\begin{aligned}
 P_{\mathbf{w}}(x) &= \int P_{\mathbf{w}}(x|z) P(z) \, dz \\
 &= \boxed{C} \int \exp \left\{ -\frac{\|x - G_{\mathbf{w}}(z)\|^2}{2} \right\} \exp \left\{ -\frac{\|z\|^2}{2} \right\} \, dz \\
 &\quad \swarrow \text{some constant} \\
 &= C \int \exp \left\{ -\frac{\|x - G_{\mathbf{w}}(z)\|^2 + \|z\|^2}{2} \right\} \, dz
 \end{aligned}$$

$G_{\mathbf{w}}(z)$ is a *complex model*, e.g., NN $\rightsquigarrow$ this is *not* a *tractable task!*

MLE Training: Formulating Challenge

Moral of Story

Using a *probabilistic generator*, we can *approximate* data distribution *very well*

↳ It's however *not tractable to train* the model by explicit *MLE*!

Let's formulate the *training challenge*: we *have access to*

- *latent* distribution $P(z)$, and
- *conditional generative* model $P(x|z)$

We intend to compute the *marginal generative* model which gives us *likelihood*

$$P(x) = \int P(x|z)P(z) dz$$

We can overcome this *challenge* using *variational inference framework*